

Cybersecurity

Career Prep and Certifications

Lesson 23.1



Learning Objectives

By the end of today's class, you will be able to:

- 1 Understand the importance of career and job-search preparation.
- 2 Understand the value of certifications in your job searches and career development.
- 3 Map certification road maps based on your specific experience level and field of interest.
- 4 Use CompTIA's CertMaster tool to begin preparing for the Security+ exam.
- 5 Understand performance-based questions (PBQs) and how to answer them.



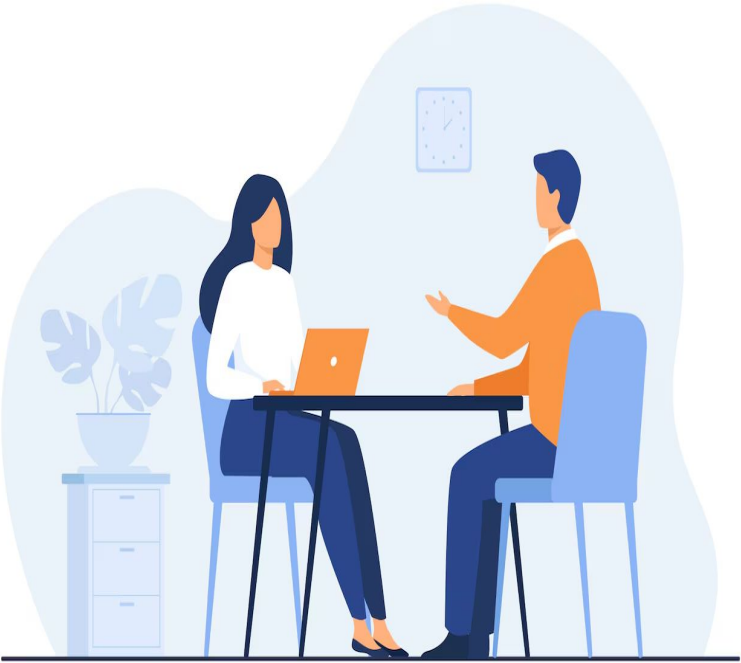
Welcome

Career Prep Week

Preparing for a New Career in Cybersecurity

Over the past several weeks, we've covered many different career paths and opportunities within cyber.

As the demand for cybersecurity professionals grows, employers want qualified individuals to fill an increasing number of roles.



This Week

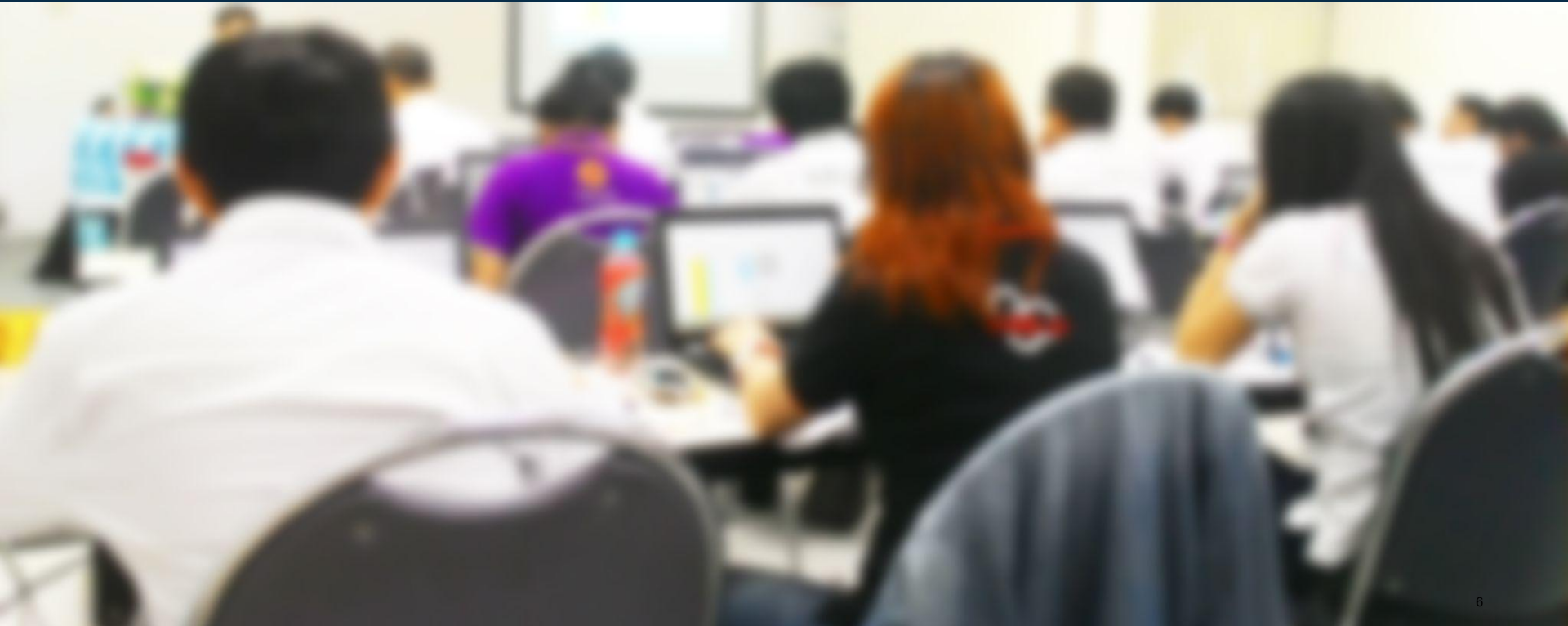
Day 1

We'll cover certifications, with a focus on Security+, and some strategies for taking certification tests.

Day 2

We will focus on mock interviews.

An important first step to landing a job or furthering your career is to demonstrate that you can do the tasks your job requires. One good way to do this is with a **certification exam**.



Certifications by the Numbers

As of April 2020...



44

organizations
issue over

300

cybersecurity
certifications for

25

specializations
and paths.

This Week

We'll also dive deeper into Security+.

This course comes with a voucher for the Security+ exam, so we'll spend the most time on this certification.

We'll cover:

- Test preparation tips
- The CompTIA CertMaster study tool
- PBQs
- Domains on the exam not covered in this course's curriculum





Information Security **Certifications**

InfoSec Certifications

Certifications can give you an advantage when applying for your first cybersecurity job, and they provide the following benefits:

01

Further Education



02

Networking



03

Career Advancement



InfoSec Certifications

01

Education

You learn new skills while preparing for a certification, even before the certification is formally awarded. While achieving credentials is often the objective, the material learned while studying for the exam is also valuable to job performance.



InfoSec Certifications

02

Networking

Many certifications have national and local organizations that host meetings, conferences, seminars, and social events. These can be opportunities to network with peers.

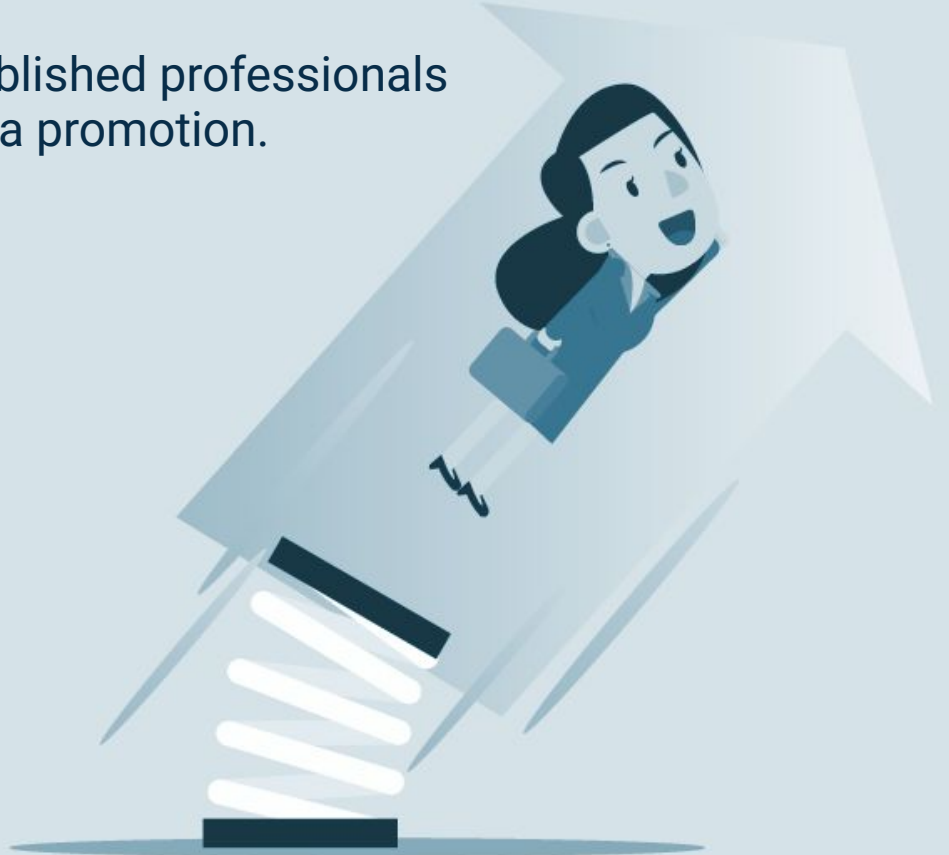


InfoSec Certifications

03

Career advancement

Certifications put more established professionals in a stronger position to get a promotion.





Cybersecurity professionals may pursue different types of certificates depending upon the **stage** of their career.

Types of InfoSec Certifications

We can break certifications into three categories:



Beginner InfoSec Certifications

These are the first certifications individuals new to the field should obtain.



Advanced InfoSec Certifications

These should be obtained after working in the industry for several years.



Specialized InfoSec Certifications

These focus on a specific domain and should be obtained after working in the industry for several years.

Beginner Certifications

Entry level and foundational towards more advanced certificates

These certifications typically **do not have** minimum work requirements or prerequisite courses.

They typically cover a broad range of subjects.



Advanced Certifications

Many of these certifications focus on security management.

These certifications **typically** •————
have minimum work
requirements and
prerequisite courses.



They typically cover a broad •————
range of subjects.



Specialized Certifications

Security professionals obtain these after working in the industry for several years.

These certifications **typically** have minimum work requirements and prerequisite courses.



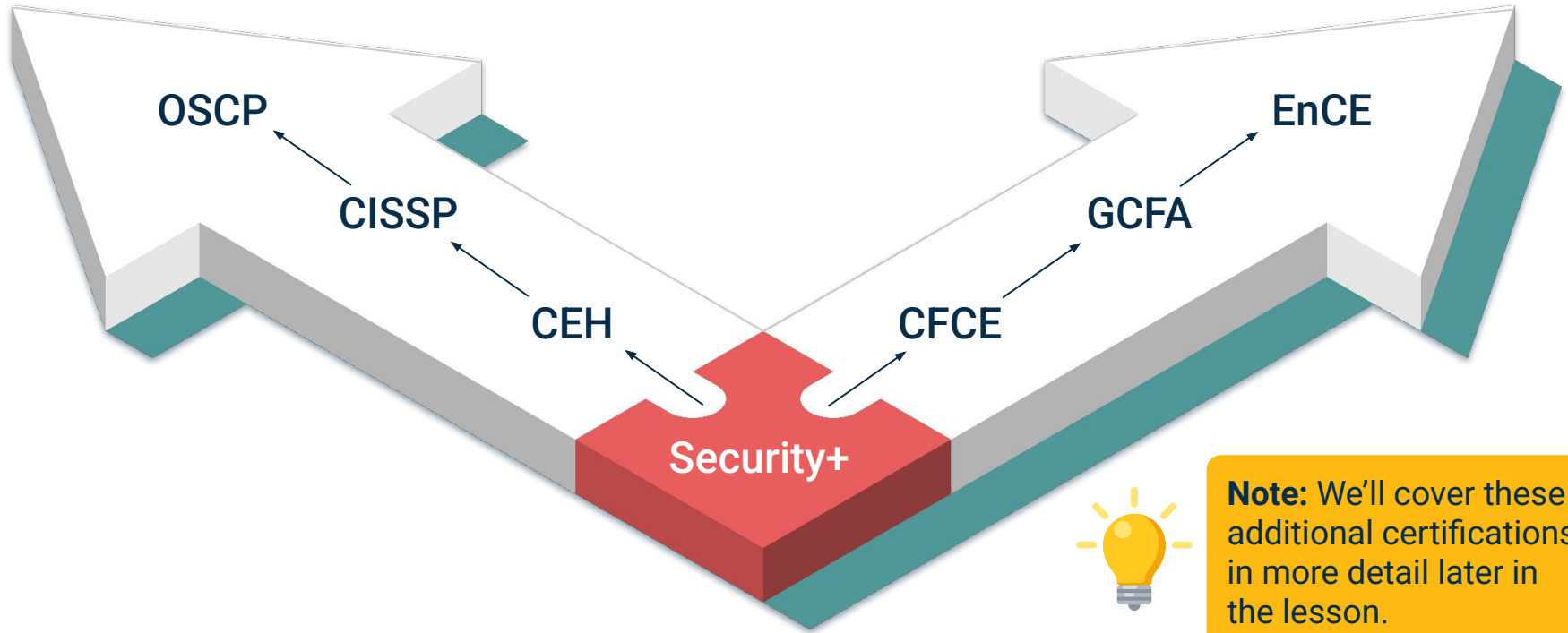
Cover **specific** subjects and can be vendor or non-vendor specific.



Certification Pathways

Penetration testers
may follow this path:

Forensic investigators
may follow this path:



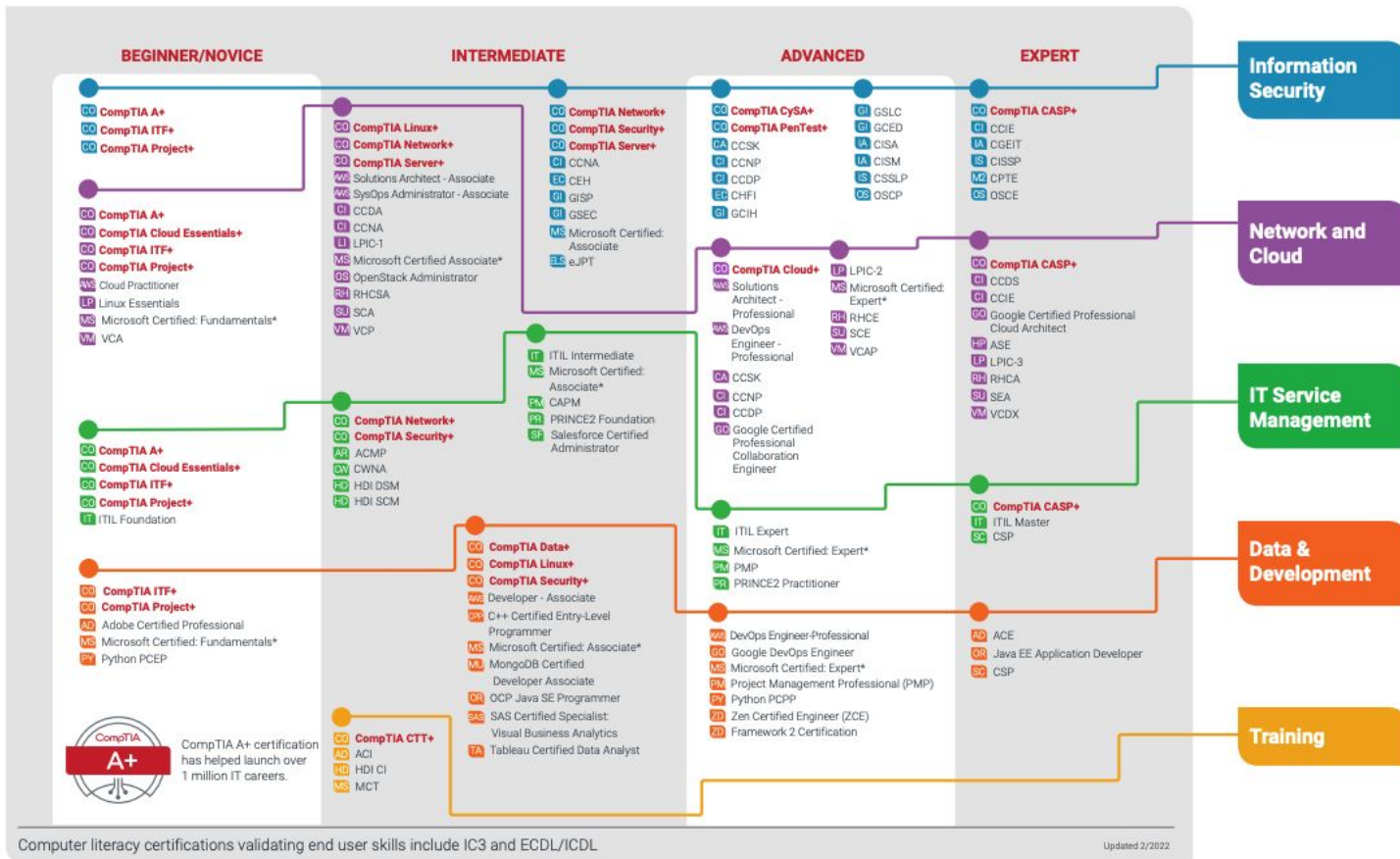
Note: We'll cover these additional certifications in more detail later in the lesson.

IT Certification Roadmap

Explore the possibilities with the CompTIA Interactive IT Roadmap at:
CompTIA.org/CertsRoadmap

CompTIA

Certifications validate expertise in your chosen career.



*Microsoft provides three certification paths. Please visit Microsoft's webpage for a full list of their offerings: <https://bit.ly/3tJYm8Z>



Activity:

Certifications and Careers

In the next activity, you will use job search websites to research InfoSec careers and certifications.

- You are tasked with finding five information security careers and the desired or required certifications that align with each.
- You will add your findings to the spreadsheet provided.

Suggested Time:
20 Minutes

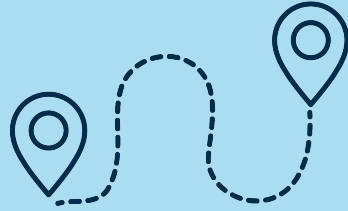


Time's up!
Let's review



Questions?





Certification **Pathways**

Beginner Certifications

While Security+ will likely be the first certification you'll work toward, several other beginner InfoSec certifications don't require experience and cover many domain areas.





Certification Pathways

Beginner: Certified Ethical Hacker (CEH)



Offered by **EC Council**

- Focuses on penetration testing
- Includes a range of topics for InfoSec professionals
- Offers an advanced version: CEH Practical
 - Hands-on environment



Certification Pathways

Beginner: GIAC Security Essentials (GSEC)

Offered by **Global Information Assurance Certification (GIAC)**

- Covers a broad spectrum of topics ranging from active defense to cryptography





Certification Pathways

Beginner: Systems Security Certified Professional (SSCP)

Offered by **ISC2**



SSCP®

Systems Security
Certified Practitioner

- SSCP covers security best practices for setting up, monitoring, and administering IT infrastructure.
- While SSCP is one of the entry-level certifications offered by ISC2, it does request one year of professional cybersecurity experience.

Many cyber professionals advance their careers by obtaining **advanced** and specialized certifications.





Certification Pathways

Advanced: Certified Information Systems Security Professional (CISSP)

Offered by **ISC2**

- It covers a wide variety of cybersecurity topics and is one of the most popular certifications in information security.



Certified Information
Systems Security Professional



Certification Pathways

Advanced: Certified Information Security Manager (CISM)



Certified Information
Security Manager®

An ISACA® Certification

Offered by **ISACA**

- It covers information security management topics.

Specialized Certifications

There are hundreds of **specialized certifications** for those interested in mastering a specific cybersecurity domain.





Certification Pathways

Specialized: Offensive Security Certified Pentester (OSCP)

Offered by **Offensive Security**

- It is a hands-on penetration testing exam consisting of **two parts**: a 24-hour penetration testing exam and a documentation report due 24 hours after the exam.





Certification Pathways

Specialized: Certified Information Privacy Professional (CIPP)



Offered by **IAPP**

- It is considered the "**gold standard**" for those working in privacy.



Certification Pathways

Specialized: EnCase Certified Examiner (EnCE)

Offered by **OpenText**

- Vendor-specific specialized certification for computer forensics professionals
- Used by law enforcement





Activity:

Find a Certification Pathway

In this activity, you will research several advanced and specialized certifications.

Suggested Time:
20 Minutes





Time's up!
Let's review



Questions?





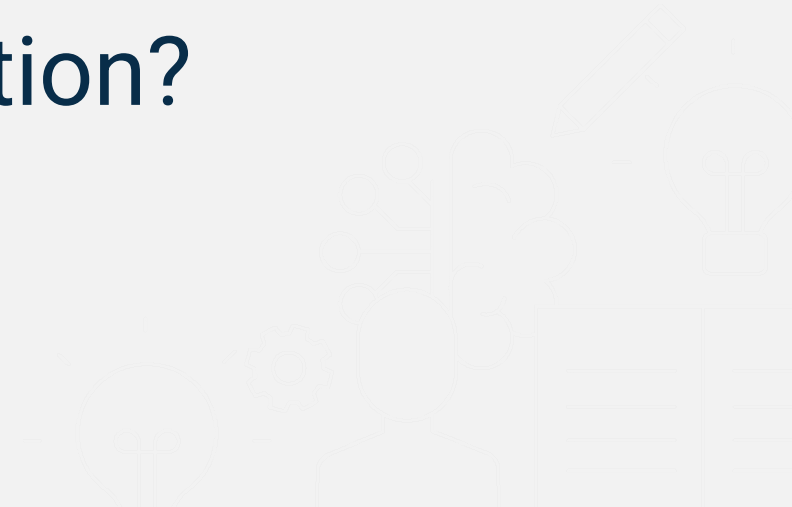
Break
15 mins



Introduction to **Security+**



What is the **Security+**
certification?





CompTIA

"Security+ is the first security certification IT professionals should earn."

"It establishes the core knowledge required of any cybersecurity role and provides a springboard to intermediate-level cybersecurity jobs."

"Security+ incorporates best practices in hands-on troubleshooting to ensure security professionals have practical security problem-solving skills."

"Cybersecurity professionals with Security+ know how to address security incidents—not just identify them."

Introduction to Security+ Certification

Some jobs that may require the **Security+** certification:

01

Security Architect

02

Security Engineer

03

Security Consultant

04

Security Specialist

05

InfoSec Analyst

Introduction to Security+ Certification

What domains does **Security+** cover?

01

Attacks, Threats, and Vulnerabilities

02

Architecture and Design

03

Implementation

04

Operations and Incident Response

05

Governance, Risk, and Compliance

Introduction to Security+ Certification

Security+ is obtained by passing the CompTIA-administered Security+ exam.

01

There are 90 multiple-choice and performance-based questions.

02

The exam lasts 90 minutes.

03

The cost of the exam is \$392.

04

The exam is vendor-neutral.

Preparation Tips

While there are many books, online resources, study guides, and apps to help prepare for the exam, CompTIA provides the most up-to-date resources for exam prep.



Preparation Tips

Design a structured study plan and stick with it. Block out at least several weeks to study for the exam.



Preparation Tips

Early on, attempt to determine which domains are the most challenging, and focus your studies there.



Preparation Tips

Create study guides and flashcards of common terms and acronyms.
Find video content that explains confusing subjects.



Preparation Tips

Many blogs, forums (e.g., Reddit), and wiki pages are dedicated to preparing for the exam.



Test-Taking Tips

Take care of yourself.

- Arrive early, well-rested, fed, hydrated, and relaxed.

Pay attention.

- Read each question and answer option twice before deciding on your answer.
- Look for keywords in the questions, such as “best,” “most,” or “least.”
- Always stay aware of how much time remains.

If you're unsure of an answer, guess.

- Eliminate the answers you know are wrong, and select from the remaining answer options.
- You'll receive the same penalty for an incorrect answer as for an unanswered question, so always make a best guess.



Security+ CertMaster

Security+ CertMaster

You'll be provided with access to the CertMaster Practice tool. This comes with:

01

Quick knowledge assessment

02

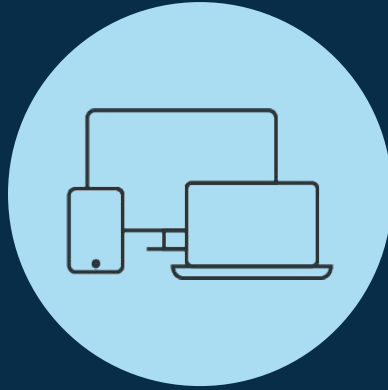
Adaptive learning that reinforces existing and new knowledge

03

Personalized feedback

04

Real-time learning analytics



Instructor Demonstration

Security+ CertMaster

Begin Setting Up the Free Trial

Go to <https://www.comptia.org/training/certmaster-practice/security>

CertMaster Practice



CertMaster Practice is an intelligent online course that helps you learn fast and remember information long-term as you prepare for the exam. It quickly assesses what you know and then focuses on filling your knowledge gaps. It's a great addition to your exam prep plan and guides you to mastery of the curriculum.



Sign Up for Your Free Trial

Complete this form to register for the trial. Once you have registered, login to get started learning!

☐ I agree to the CompTIA CertMaster [License Terms](#)

☐ I agree to the [Terms of Use](#) & [Privacy statement](#).

START YOUR FREE TRIAL NOW

Review the Learning Module

The free trial only has one module. The full CertMaster Tool will cover all domains of the exam.

your modules

CertMaster Practice for Security+ (Exam SY0-701) - Trial

▼ Security+ Sample Learning Module

Module	Progress
Security+ Sample Learning Module begin learning 1 hr 3 min	

▶ Security+ Sample Practice Test

Answering Questions

Click an answer twice to fill in the circle if you're certain it is the correct answer.

QUESTION



As part of enhancing its data protection strategy, a corporation's IT manager aims to ensure defense-in-depth by integrating a technical control alongside existing managerial and operational controls. Which measure BEST exemplifies a technical security control according to the classification scheme?

ANSWER

- ☐ Conducting employee cybersecurity training
- ☐ Installing a building access control system
- ☐ Implementing a risk identification tool
- ☒ I AM **SURE**
Setting up a network intrusion detection system
- ☐ I DON'T KNOW YET

submit

Answering Questions

If you are unsure of the answer, you can click multiple answers once.

QUESTION



As part of enhancing its data protection strategy, a corporation's IT manager aims to ensure defense-in-depth by integrating a technical control alongside existing managerial and operational controls. Which measure BEST exemplifies a technical security control according to the classification scheme?

ANSWER

- ☐ Conducting employee cybersecurity training
- ☒ I AM **UNSURE**
Installing a building access control system
- ☐ Implementing a risk identification tool
- ☒ I AM **UNSURE**
Setting up a network intrusion detection system
- ☐ I DON'T KNOW YET

submit

Submitting Answers

The tool will let you know if any of your answers are correct.

QUESTION



As part of enhancing its data protection strategy, a corporation's IT manager aims to ensure defense-in-depth by integrating a technical control alongside existing managerial and operational controls. Which measure BEST exemplifies a technical security control according to the classification scheme?

ANSWER

- ☐ Conducting employee cybersecurity training
- ☒ YOU WERE **UNSURE** AND **1 IS CORRECT**
Installing a building access control system
- ☐ Implementing a risk identification tool
- ☒ YOU WERE **UNSURE** AND **1 IS CORRECT**
Setting up a network intrusion detection system
- ☐ I DON'T KNOW YET

next question >>

Reviewing Answers

The tool will let you know which answer is correct.

LEARN

QUESTION

REVIEWING 4 OF 4



As part of enhancing its data protection strategy, a corporation's IT manager aims to ensure defense-in-depth by integrating a technical control alongside existing managerial and operational controls. Which measure BEST exemplifies a technical security control according to the classification scheme?

ANSWER

YOU CHOSE 2



Conducting employee cybersecurity training



YOU WERE UNSURE AND INCORRECT
Installing a building access control system



Implementing a risk identification tool



YOU WERE UNSURE AND CORRECT
Setting up a network intrusion detection system



I DON'T KNOW YET

Reviewing Answers

The tool will let you know which answer is correct.

LEARN

QUESTION

REVIEWING 4 OF 4



As part of enhancing its data protection strategy, a corporation's IT manager aims to ensure defense-in-depth by integrating a technical control alongside existing managerial and operational controls. Which measure BEST exemplifies a technical security control according to the classification scheme?

ANSWER

YOU CHOSE 2



Conducting employee cybersecurity training



YOU WERE UNSURE AND INCORRECT
Installing a building access control system



Implementing a risk identification tool



YOU WERE UNSURE AND CORRECT
Setting up a network intrusion detection system



I DON'T KNOW YET

Additional Tips

While you are working on a module, you can view your current progress by clicking the arrow on the top right of your module.

Learning: Security+ Sample Learning Module



time

estimated time to complete

1 hr 1 min

accuracy 



70%

progress

1  mastered 0  incorrect
5  in progress 22 ☐ not seen
28 total questions

message

Please answer with honesty. If you are unsure or don't know, say so.

After you complete the module, you can view a summary of the completed modules.

From the dashboard, select **Course Report**.

course overview

	you	all learners
progress	100%	24%
time spent	44min	47min
starting knowledge	71%	37%
refreshers taken	0	0.1

[course report](#)



Additional Tips

Additionally, the **Course Report** page shows the most challenging questions, which you can return to and re-answer

CompTIA CertMaster Practice for Security+ SY0-701 Trial

[back to dashboard](#)

Course Average

	you	all learners
progress	43%	24%
time spent	1hr 11min	47min
starting knowledge	25%	37%
refreshers taken	0	0.1

Course Progress



- 0 Modules Not Started
- 1 Modules in Progress
- 0 Modules Completed

Most Challenging Questions

- A malicious pen tester ...
- Which of the box testi...
- A security consultant ...



Activity:

Security+ CertMaster

In this activity, you will use the Security+ CertMaster study trial to complete the sample test module.

Suggested Time:
10 Minutes





Time's up!
Let's review



Questions?





Security+ PBQs

Security+ PBQs

The Security+ exam has **two types** of questions: multiple choice and performance-based questions (PBQs).

01

PBQs present a simulated environment, such as a network, firewall, or terminal.

02

Since the question environment is simulated and not live, there may be some limitations compared to a real-world environment.

03

PBQs are often the **first questions** on the Security+ exam.

PBQs

Let's explore Security+ PBQ questions.

Welcome to the CompTIA Example Simulation

Read the question carefully, follow the instructions, and then click the submit button when you have finished. You will receive a numeric score once you have submitted a response.

Submit

TEST QUESTION



After experiencing attacks on its servers, Company A hired a cybersecurity analyst to configure a DMZ and increase security measures.

Shortly after the network was reconfigured, an assistant on the 2nd floor reported that one of the executives could not access the Internet (more specifically <https://compia.org>).

However, he said, they can send internal Email, use the intranet, and print on the local area network printer.

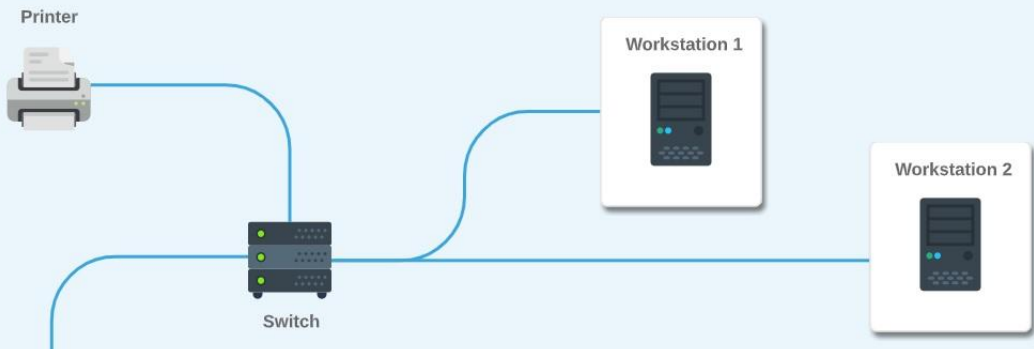


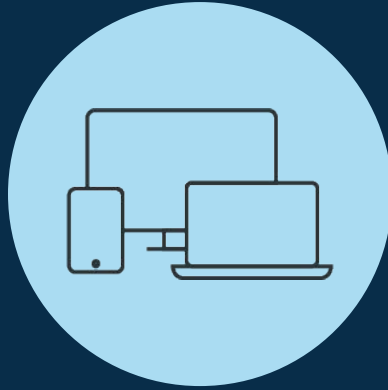
Show Question



Reset All Answers

Floor 2 - Executive Offices





Instructor Demonstration

Security+ PBQs



Activity:

Security+ PBQs

In this activity, you will work through several Security+ PBQs.

- You are tasked with completing all of the questions in the allotted time.
- Be sure to read all the instructions carefully before completing the activity.

Suggested Time:
15 Minutes

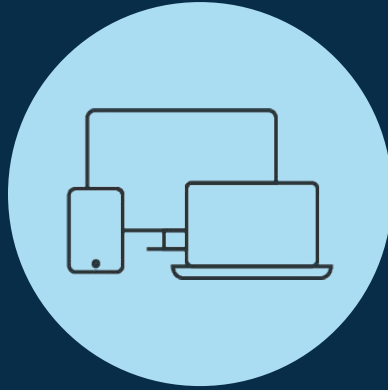


Time's up!
Let's review



Questions?





Instructor Demonstration

Security+ PBQs



Let's recap

Module Recap

We've examined a lot of information and done the following:

01

Performed mobile forensic analysis and compiled details of Peter's iPhone

02

Searched through numerous files and directories on the iPhone image

03

Tagged and categorized evidence relevant to the case

04

Created custom tags

Module Recap

We've examined a lot of information and done the following:

05

Extracted data for offline analysis using the export function

06

Examined and documented Peter's emails

07

Examined and documented Peter's SMS messages

Class Wrap-Up

Key takeaways from this lesson:

01

Digital forensics is a field dedicated to identifying, extracting, preserving, and reporting against information obtained from computer and network systems.

02

Digital forensics relies upon the expertise of examiners to analyze and interpret data retrieved by trusted forensics examination tools.

03

Investigative teams may be spread across several time zones, so it's important to follow a baseline timezone as indicated by case file.



You'll continue your report
and work on another
bonus assignment for this
week's Challenge.



Questions?





Next class

Cybersecurity Interviewing



Thank you